Lab 4: Alpha Spectroscopy and Range Measurement

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Abstract

1 Introduction and Background

A key metric of interest for radiation in matter is range, that is, the depth along the initial direction which a particle is expected, on average, to penetrate through a material. For a given type of particle, this range is further highly dependent on an incident particle's energy and the electron/neutron densities of the medium.[1] This laboratory investigates the range of alpha particles, as emitted by ^{241}Am and ^{210}Po, through aluminum metal, Mylar[1] plastic, and air. More broadly, the energy loss phenomena of alpha particles through different materials, including energy straggling, and the effects of geometric attenuation corresponding to solid angle, are also investigated.

1.1 Charged Particle Energy Loss

2 Experimental Procedure

In this section, we detail the equipment used and experiments performed in this laboratory. Further equipment details may be found in Appendix A

2.1 Equipment

In this laboratory, we employed

[1] While material names are usually kept in lowercase, "Mylar" is a brand name, and remains accordingly capitalized in this report.

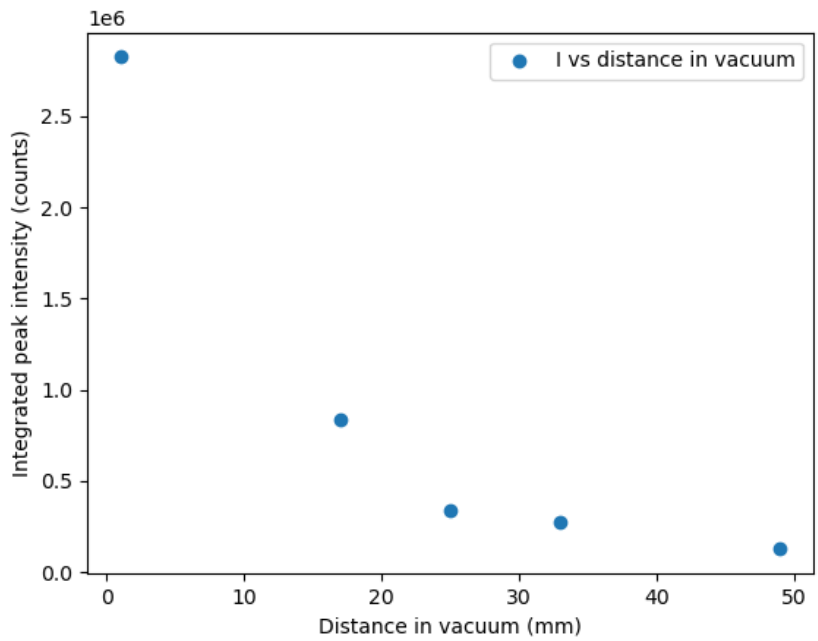

Figure 1: The integrated peak intensity of the alpha source after attenuation through varying distances in vacuum.

3 Experimental Results

In Fig. 1, the integrated count rate for the alpha source is shown after attenuation through different distances in vacuum. However, to does not account for the geometric attenuation, one must first normalize by solid angle Ω, using the following formula[2]:

$$\Omega = 2\pi \left(1 - \frac{d}{\sqrt{d^2 + a^2}}\right) \tag{1}$$

where d is the distance of the (point) source from the detector, and a the diameter of the detector, a sufficiently long cylinder. This formula assumes the point source is centered on the cylindrical detector's axis.

If the source can "see" $\frac{\Omega}{4\pi}$ of the detector, then dividing by this fraction yields an intensity normalized by solid angle. Fig. **??** shows the integrated count rate for distances through vacuum, normalized by solid angle, compared to the original rates.

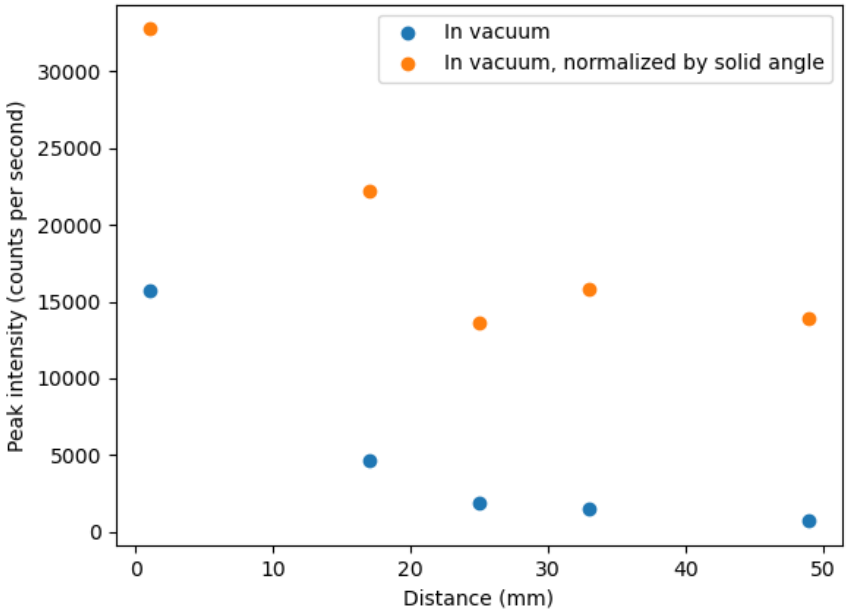

Figure 2: The integrated peak intensity of the alpha source after attenuation through varying distances in vacuum, both raw and after normalizing by visible solid angle.

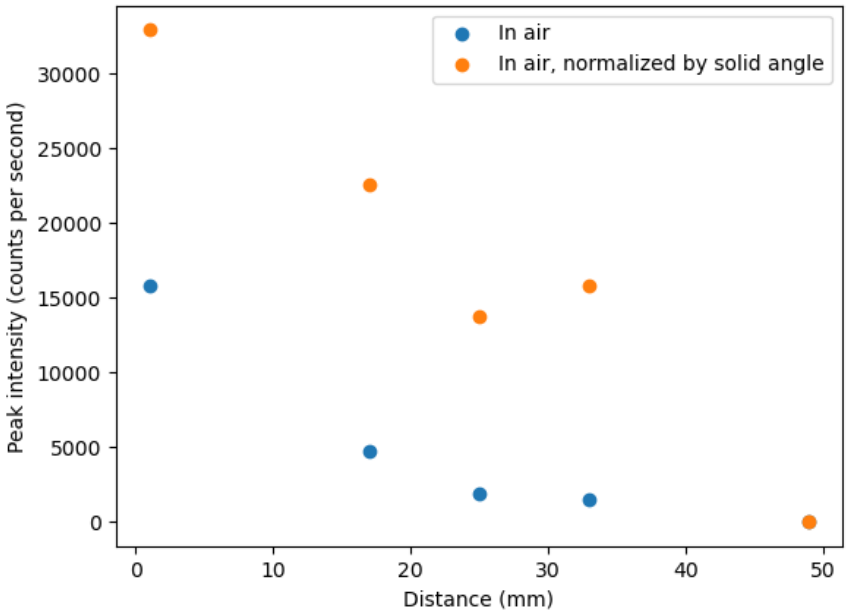

Figure 3: The integrated peak intensity of the alpha source after attenuation through varying distances in air, both raw and after normalizing by visible solid angle.

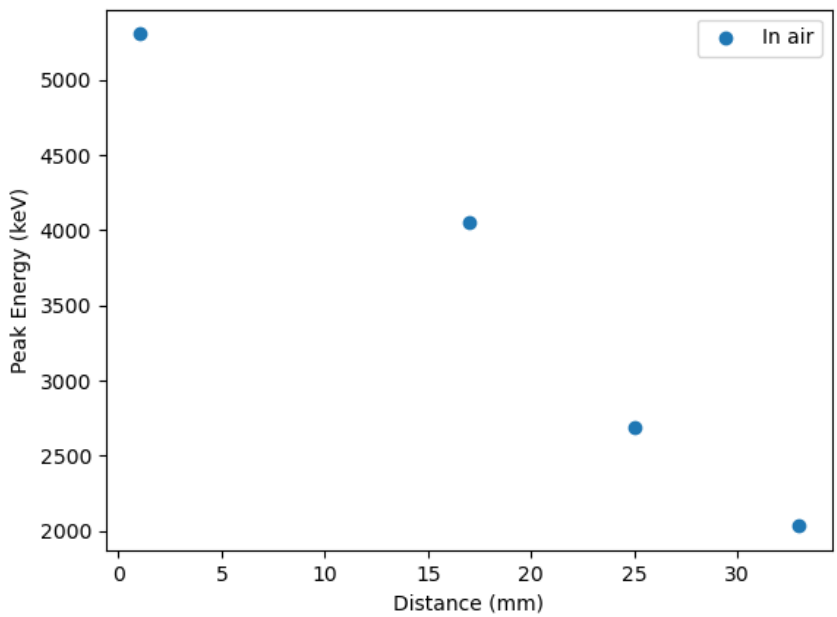

Figure 4: The peak energy after attenuation through different distances in air.

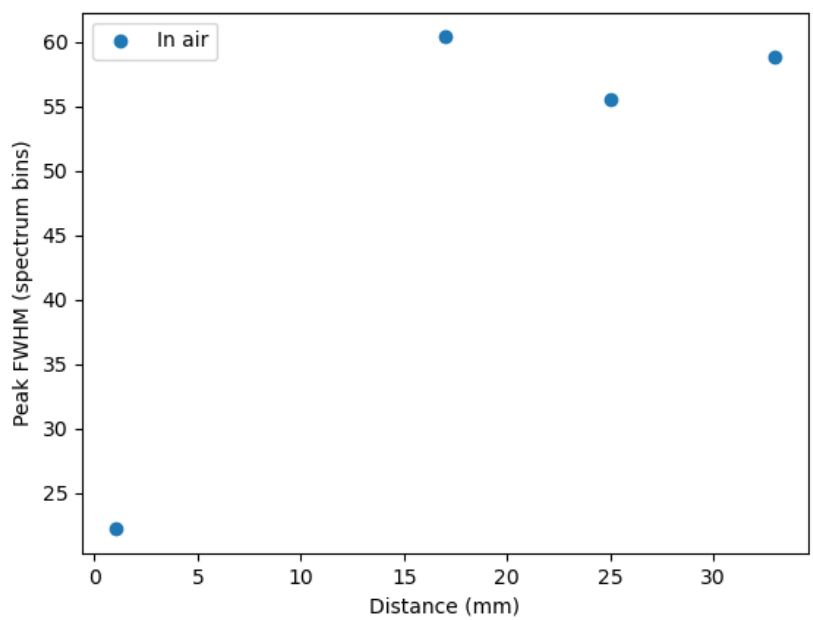

Figure 5: The full width half maximum of peak energy after attenuation through different distances in air.

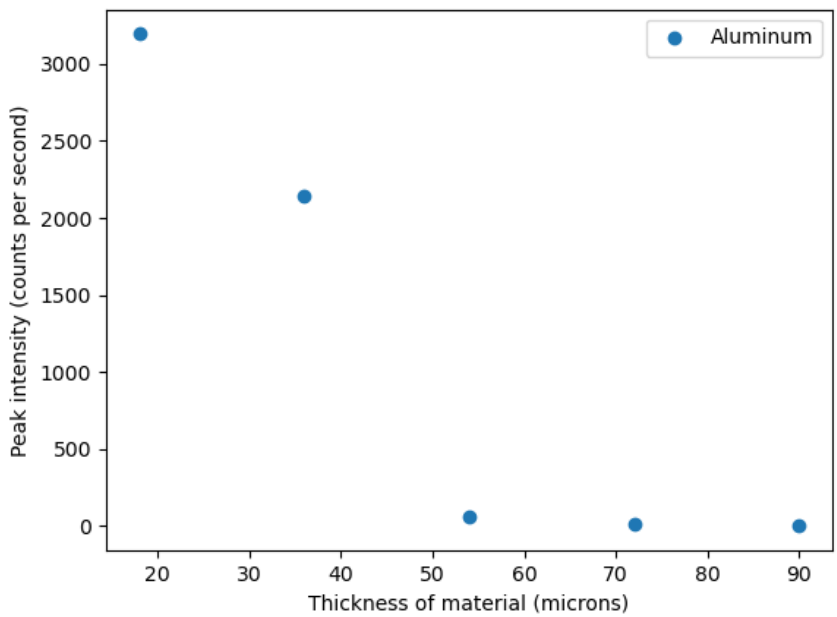

Figure 6: The integrated peak intensity after attenuation through varying thicknesses of aluminum.

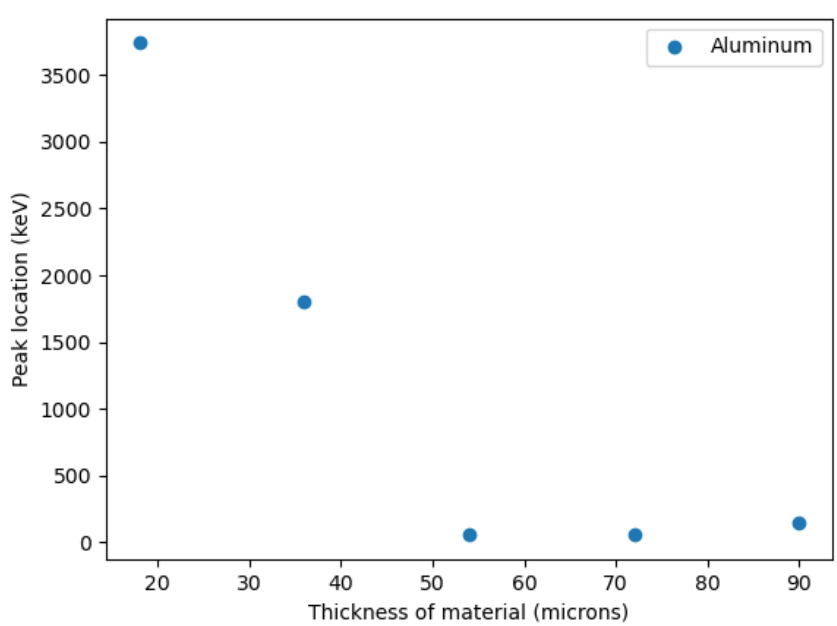

Figure 7: The peak energy after attenuation through varying thicknesses of aluminum.

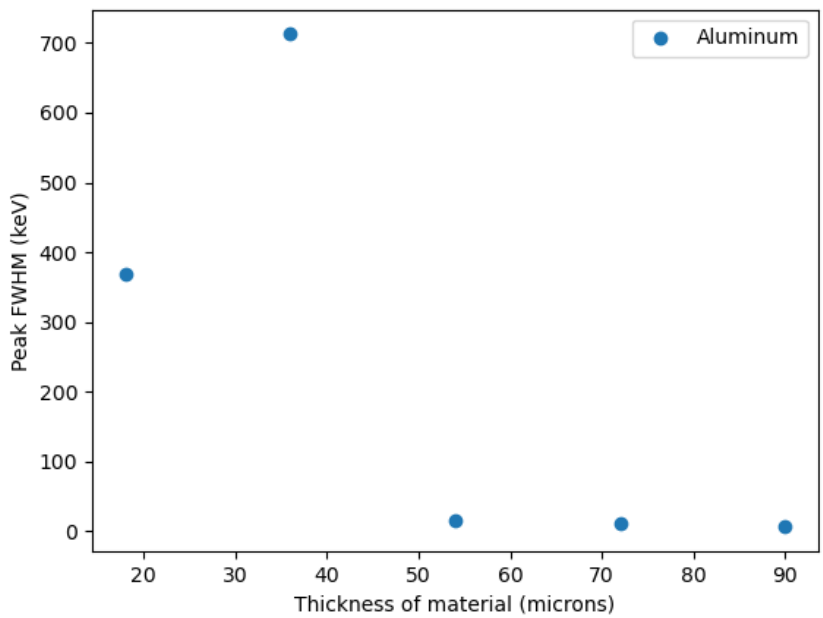

Figure 8: The full width half maximum of peak energy after attenuation through varying thicknesses of aluminum.

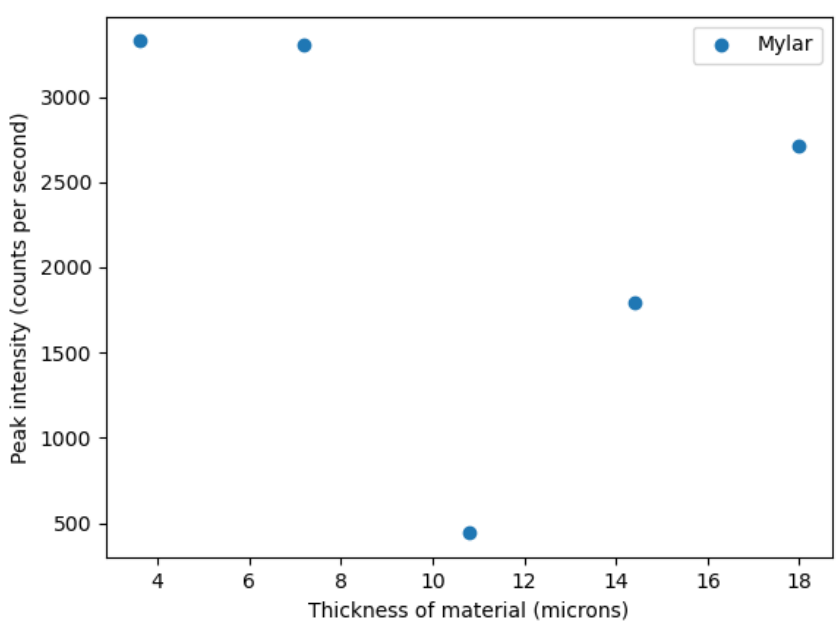

Figure 9: The integrated peak intensity after attenuation through varying thicknesses of Mylar.

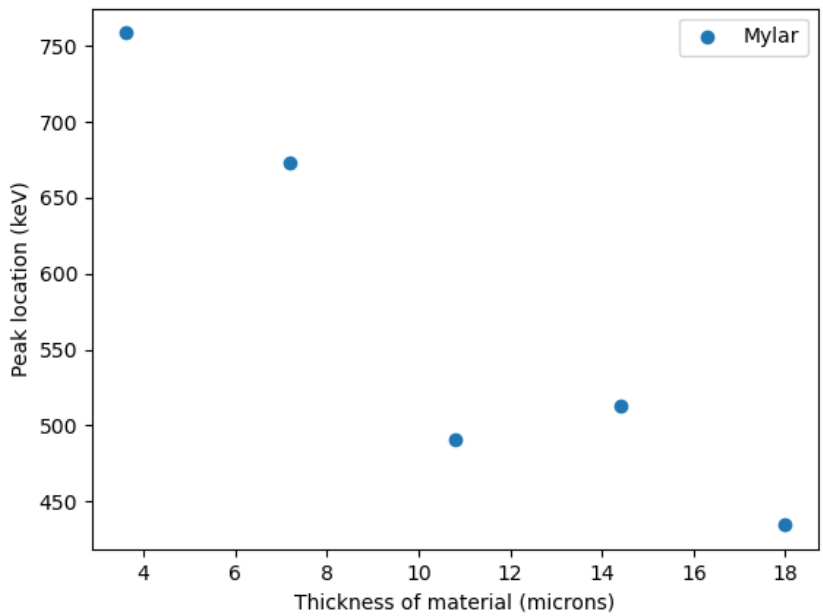

Figure 10: The peak energy after attenuation through varying thicknesses of Mylar.

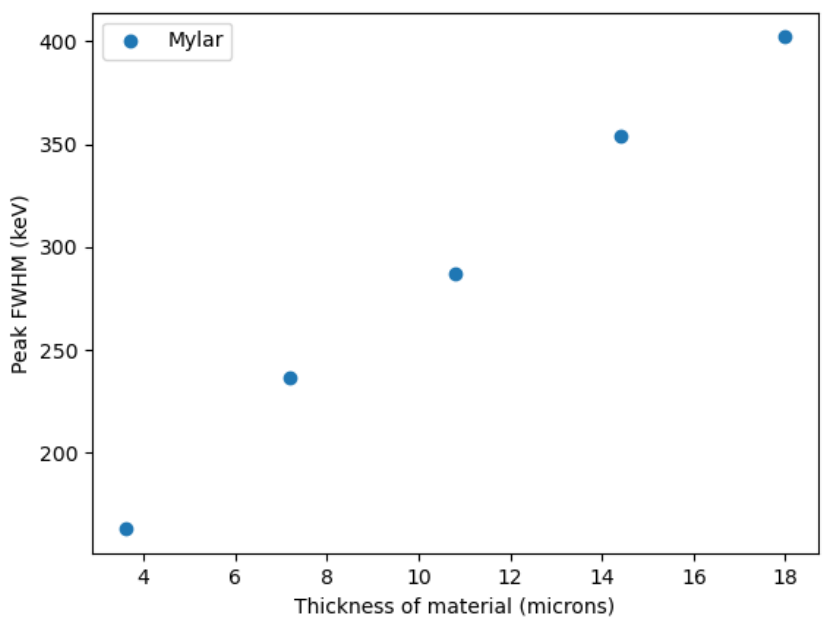

Figure 11: The full width half maximum of peak energy after attenuation through varying thicknesses of Mylar.

4 Discussion

5 Conclusions

References

[1] D. o. NPRE, "Lab_4_manual," University of Illinois at Urbana-Champaign, Urbana, IL, Lab Manual 4, 2025.

[2] G. F. Knoll, *Radiation detection and measurement*, 4th ed. New York: Wiley, 2010, 830 pp., ISBN: 978-0-470-13148-0.

A Equipment Details

In order to better replicate experiments and identify issues, the details of each equipment piece are recorded in Table 1.

Further details of equipment are listed in Table 1. Where a detail could not be identified for a given device, it is replaced with "N/A".

Table 1: Details of the equipment used in this laboratory.

Item	Manufacturer	Model	ID Type	ID Number
Module rack	Canberra (now Mirion)[a]	Model 2000	Inventory #	NE759377
Geiger-Muller Detector	N/A	N/A	N/A	N/A
GM Pulse Inverter	Spectrum Techniques[b]	2017 GPI	N/A	N/A
High Voltage Power Supply	Caen[d]	N1470AL	Inventory #	P10G96054
Computer	Dell[g]	Precision 360	Eng. IT ID #	EWS 20426

a Mirion Technologies, Inc.: 1218 Menlo Drive, Atlanta, GA; Phone: 770.432.2744; https://ir.mirion.com/

b Spectrum Techniques: 106 Union Valley Rd, Oak Ridge, TN 37830; Phone: 865.482.9937; https://www.spectrumtechniques.com/

c Eckert & Ziegler Isotope Products: 24937 Avenue Tibbitts, Valencia, CA 91355; Phone: +1 866 476-9767; https://isotopeproducts.com/

d Caen Sp.A.: Via della Vetraia, 11, 55049 Viareggio LU - Italy; Phone: +39 0584 388398; https://www.caen.it/

e Keysight Technologies: 1900 Garden of the Gods Road, Colorado Springs, CO 80907-3423 United States; Phone: 1 800 829-4444; https://www.keysight.com

f Advanced Measurement Technology: 801 South Illinois Avenue, Oak Ridge, Tennessee 37830, United States; Phone: +1.865.482.4411; Fax: +1.865.483.0396; https://www.ortec-online.com; Email: ortec.info@ametek.com

g Dell Technologies: Dell Technologies 1 Dell Way Round Rock, TX 78664.; Phone: 1-877-275-3355; https://www.dell.com/en-us/lp/contact-us